can be operated at a high temperature. It provides a high intensity light.

a. Advantages

1. It is smaller in size.
2. It does not need any ballast.
3. Good colours can be obtained.
4. Excellent optical control.
5. It gives same output throughout life
6. It has long life

b. Disadvantages

1. During maintenance the handling of lamp is very difficult.
2. Radiation is more which heats the surroundings also.
3. Operating temperature is high which affects its life.

2.9 LED lamp



Fig. 2.14 **LED Lamp**

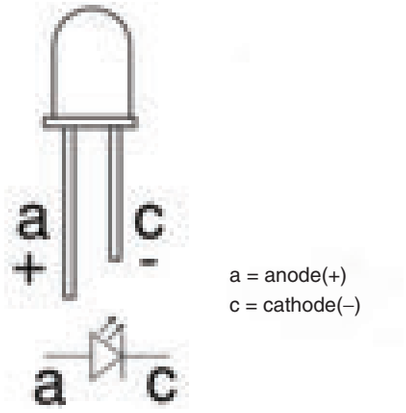


Fig. 2.15 **Line diagram of LED lamp**

The construction of LED Lamp is shown in figure 2.14. An LED lamp is a light-emitting diode (LED) product that is assembled into a lamp (light or bulb) for use in lighting fixtures. The line diagram of a LED lamp is shown in figure 2.15.

LED lamps have a lifespan and electrical efficiency which are several times greater than incandescent lamps, and are significantly more efficient than most fluorescent lamps. General purpose lighting needs white light. LEDs emit light in a very narrow band of wavelengths. To emit white light from LEDs require either mixing light from LEDs of various colours, or using phosphorous to convert some of the light to other colours.

Basic components for LED lighting are:

1. LED
2. Driver (power conversion device)
3. Control devices (dimming controls, colour mixing controls)
4. Fixture

a. LED driver

Required by the LED system to convert a system voltage into power. The driver also regulates power delivered to LEDs to counter any fluctuations in system conditions. Drivers also isolate the LED system from the high voltage system to reduce shock hazards and make the lighting system safer.

b. Advantages

1. It has a long life.
2. It reduces energy costs.
3. It reduces maintenance costs
4. LEDs produce very little heat.

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